

Predictive Credit Risk Assessment

Farzana zahid, Israt Jahan, Ibteshum Khaled

Department of Computer Science, Ahsanullah University Of Science and Technology

Email: isratjahan7908@gmail.com

Abstract—Credit risk management is a fundamental component in financial institutions. This research applies data mining techniques for credit risk assessment using the Weka tool. By analyzing customer datasets, the study explores classification, association rule mining, and clustering methods to evaluate their performance in predicting creditworthiness.

I. INTRODUCTION

Credit risk management ensures the stability of financial systems. Traditional evaluation methods rely on human judgment and are prone to errors. With the availability of customer and financial datasets, data mining has emerged as a reliable way of identifying risk patterns. The purpose of this study is to utilize Weka-based techniques to classify, cluster, and analyze customer attributes for credit risk prediction.

A. Problem Statement

Financial institutions face difficulties in minimizing credit losses while maximizing profitability. The problem is determining how to apply data mining to classify customers into high-risk, medium-risk, and low-risk groups effectively.

B. Objective

- Develop a Predictive Model: Build a machine learning model to classify loan applicants as "Good Credit" or "Bad Credit".
- Customer Segmentation: Discover distinct customer groups with similar risk profiles using unsupervised learning techniques.
- Pattern Discovery: Mine association rules to identify key relationships between applicant characteristics and credit outcomes.
- Data Preprocessing: Implement comprehensive data cleaning and transformation processes
- Model Comparison: Evaluate multiple supervised learning algorithms to identify the best performer
- Business Intelligence: Generate actionable insights for loan officers and risk managers
- System Implementation: Create a deployable Java application using WEKA API for real-time credit scoring

II. REVIEW OF RELATED RESEARCH

Credit risk assessment is a crucial task for financial institutions to evaluate borrower default. Machine learning and data mining techniques implemented in Java and Weka have been widely applied.

Škegro et al. [1] utilized the German credit dataset from the UCI repository and developed a decision tree model using

Weka's J.48 algorithm, highlighting the interpretation of Weka outputs for credit risk analysis.

Biju and Koshy [2] applied machine learning in Weka to predict loan defaults, analyzing customer attributes such as income, employment status, credit history, and loan approval status.

Other studies [3] explored combined supervised and unsupervised learning approaches to improve credit card profit models.

A comparison of machine learning techniques such as SVM, Neural Networks, Logistic Regression, Naive Bayes, Random Forest, and CART was conducted in [4]. Random Forest showed higher accuracy.

Fenerich et al. [5] evaluated different machine learning techniques for classifying payers in bank financing.

Yin [6] proposed a stacking ensemble machine learning model for personal credit risk assessment on P2P lending platforms.

III. LITERATURE REVIEW

Six related works were reviewed focusing on classification, clustering, and hybrid models. Each research identified strengths in accuracy but faced limitations such as dataset imbalance or limited generalizability. Due to space, they are summarized in the report body with detailed references.

IV. METHODOLOGY

The methodology followed in this research is illustrated in Fig. 1.

- 1) *Data collection*: from publicly available credit datasets.
- 2) *Preprocessing*: handling missing values, discretization, normalization.
- 3) *Classification*: Applying classification algorithms (RandomForest, j48, SMO and Naive Bayes). Running clustering (K-Means) for customer segmentation. Generating association rules for hidden pattern discovery. Performance evaluation with accuracy, precision, recall, F1-score.

V. RESULT ANALYSIS

Naïve Bayes emerged as the most balanced classifier, Random Forest performed best in ROC, clustering identified meaningful customer groups, and association rules highlighted stable patterns useful for credit risk decision-making.

Dataset and Preprocessing: about 1000 instances, 21 attributes. Here, missing values replaced with mean/mode. Duration and

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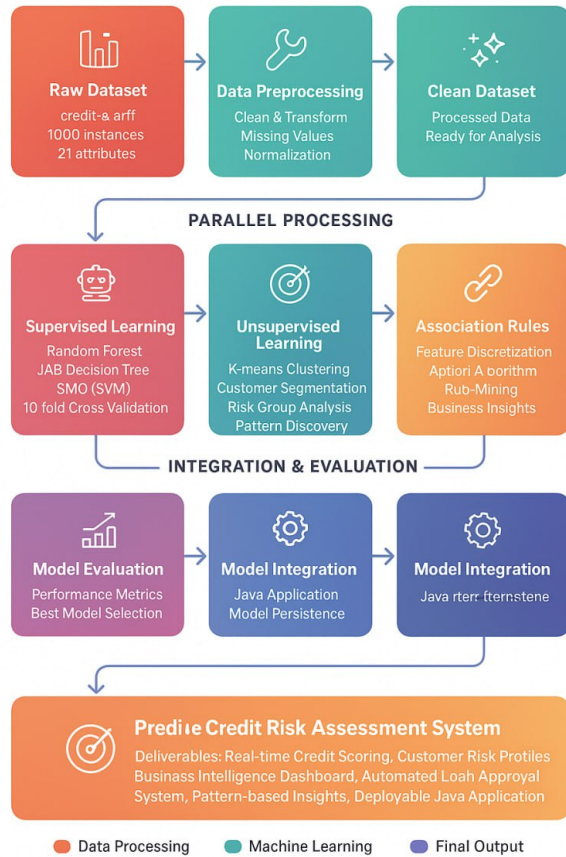


Fig. 1. Bengali Environmental Cause–Solution Framework Methodology.

credit amount positively correlated (0.625). Weak correlations observed between demographic attributes (e.g., age, dependents) and financial attributes.

Supervised Learning: Random Forest: Accuracy 75.9. Naïve Bayes: Accuracy 75.0. SMO: Accuracy 75.2. J48: Accuracy 69.7 which is the lowest.

Clustering (K-Means, k=3): Cluster 0 (401): Short loans, skilled jobs, moderate commitments. Cluster 1 (266): Larger loans, long employment history. Cluster 2 (333): Real estate owners, stable residence, older customers.

Association Rules (Apriori): Strong rules found with high confidence (0.95).

VI. LIMITATIONS

The research faced the following issues:

- Dataset imbalance requiring SMOTE resampling.
- Limited size of dataset restricting generalization.
- Difficulty in explaining complex models to non-technical stakeholders.

VII. CONCLUSION AND DISCUSSION

This work shows that Weka-based data mining can support credit risk assessment effectively. While classification models provide accurate predictions, clustering and association rules add interpretability. Future work should explore larger datasets and hybrid models to improve robustness.

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